

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – Scenario based

Course Code:	DSA0405	Course Title:	FUNDAMENTALS OF DATA SCIENCE
CO Assessed:	CO4 – Scenario based	Assessment:	Assessment Tool 1– Scenario based
Weightage:	40% of CIA	Total Marks:	100
CO4 Statement:	Analyze statistical claims using confidence intervals and p-values to perform hypothesis testing, make appropriate decisions, and interpret results in real-world contexts.		
Student Name:		Reg. No.:	
Section / Batch:		Date:	

Code of Conduct

I Y. Jayesh Ranjan - 192472201 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Y. Jayesh Ranjan

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	

Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

Questions:

Question 1: A university claims that at least 80% of its students are satisfied with online classes. A survey produces a 95% confidence interval of (74%, 79%) for the satisfaction proportion. What conclusion can be made about the university's claim?

Question 2: A mobile application developer claims that users spend an average of 30 minutes per day on the application. A sample produces a 99% confidence interval of (31.5, 35.2) minutes. Using the confidence interval, determine whether the claim is supported.

Question 3: A credit-card company claims that its fraud detection system identifies 95% of fraudulent transactions. After testing the system on a sample, the calculated p-value is 0.02. At a 5% significance level, what conclusion should the data scientist make?

Question 4: An e-commerce company introduces a new recommendation algorithm. The existing algorithm has a conversion rate of 12%. After implementation, the observed conversion rate is 15%. The hypothesis test produces a p-value of 0.03. At a significance level of 0.05, determine whether the new algorithm significantly improves conversion.

Question 5: A hospital introduces an AI-based patient scheduling system. The hospital wants to determine whether it reduces average waiting time. A hypothesis test produces a p-value of 0.08. Using a significance level of 5%, should the hospital conclude that the new system reduces waiting time?

QUESTION 1 — UNIVERSITY ONLINE CLASSES

Given

University's claim:

At least 80% of students are satisfied with online classes.

Therefore:

Claimed proportion = 80% = 0.80

Confidence level:

95%

Confidence interval:

(74%, 79%)

Step 1: State the hypotheses

$H_0: p \geq 0.80$

$H_1: p < 0.80$

Step 2: Use the confidence interval

The given 95% confidence interval is:

(74%, 79%)

Therefore:

$74\% \leq p \leq 79\%$

The university's claimed value is:

80%

Now compare the upper limit of the confidence interval with the claimed value:

$79\% < 80\%$

Therefore, **80% is not included in the confidence interval.**

In fact, the entire confidence interval is below 80%.

Step 3: Decision

Since the claimed value of 80% is outside the 95% confidence interval:

Decision: Reject H0

Step 4: Conclusion

The confidence interval ranges from **74% to 79%**, while the university claims that at least **80%** of students are satisfied.

Since even the upper limit of the interval, **79%**, is below **80%**, there is sufficient evidence that the actual proportion of satisfied students is less than 80%.

Final Answer

The university's claim is not supported.

There is sufficient evidence to conclude that **less than 80% of students are satisfied with online classes.**

QUESTION 2 — MOBILE APPLICATION USAGE

Given

Claimed average usage:

30 minutes per day

Confidence level:

99%

Confidence interval:

(31.5, 35.2) minutes

Step 1: State the hypotheses

H0: $\mu = 30$

H1: $\mu \neq 30$

Step 2: Use the confidence interval

The given 99% confidence interval is:

$$(31.5, 35.2)$$

Therefore:

$$31.5 \leq \mu \leq 35.2$$

The developer's claimed average is:

30 minutes

Now compare:

$$30 < 31.5$$

Therefore:

30 is outside the interval (31.5, 35.2).

The claimed value of 30 minutes is therefore not supported by the confidence interval.

Step 3: Decision

Since 30 minutes is outside the 99% confidence interval:

Decision: Reject H0

Step 4: Conclusion

The confidence interval ranges from **31.5 minutes to 35.2 minutes**.

The claimed average is only **30 minutes**, which is lower than the entire confidence interval.

Therefore, there is sufficient evidence that the true average usage is **greater than 30 minutes per day**.

Final Answer

The developer's claim is not supported.

The data suggest that users spend **more than 30 minutes per day**, on average, using the application.

QUESTION 3 — CREDIT-CARD FRAUD DETECTION

Given

Company's claim:

The fraud detection system identifies 95% of fraudulent transactions.

Therefore:

Claimed proportion = 95% = 0.95

p-value:

0.02

Significance level:

5%

Therefore:

$\alpha = 0.05$

Step 1: State the hypotheses

$H_0: p = 0.95$

$H_1: p \neq 0.95$

Step 2: Compare the p-value with the significance level

Given:

p-value = 0.02

and:

$\alpha = 0.05$

Compare:

$0.02 < 0.05$

The p-value is smaller than the significance level.

Step 3: Decision

Decision rule:

If p-value < α , reject H_0 .

Here:

$$0.02 < 0.05$$

Therefore:

Decision: Reject H0

Step 4: Conclusion

Since the p-value of **0.02** is less than the significance level of **0.05**, there is sufficient statistical evidence against the company's claim.

Therefore, the actual fraud-identification rate is statistically different from **95%**.

Final Answer

Reject the company's 95% claim.

At the 5% significance level, there is sufficient evidence to conclude that the fraud detection system's true identification rate **differs from 95%**.

QUESTION 4 — E-COMMERCE RECOMMENDATION ALGORITHM

Given

Existing conversion rate:

$$12\% = 0.12$$

New observed conversion rate:

$$15\% = 0.15$$

p-value:

$$0.03$$

Significance level:

$$5\% = 0.05$$

Step 1: Calculate the increase

Old conversion rate:

12%

New conversion rate:

15%

Increase:

$15\% - 12\% = 3 \text{ percentage points}$

Therefore, the observed conversion rate increased by:

3 percentage points

Step 2: State the hypotheses

The company wants to determine whether the new algorithm **improves** conversion.

Therefore:

$H_0: p = 0.12$

$H_1: p > 0.12$

Step 3: Compare the p-value with the significance level

Given:

p-value = 0.03

Significance level:

$\alpha = 0.05$

Compare:

$0.03 < 0.05$

The p-value is smaller than the significance level.

Step 4: Decision

Decision rule:

If p-value < α , reject H_0 .

Since:

$0.03 < 0.05$

Therefore:

Decision: Reject H0

Step 5: Conclusion

The conversion rate increased from:

12% to 15%

The observed increase is:

15% - 12% = 3 percentage points

Since the p-value is:

0.03

and:

0.03 < 0.05

the improvement is statistically significant at the 5% significance level.

Final Answer

The new recommendation algorithm significantly improves conversion.

There is sufficient evidence at the 5% significance level that the new algorithm increases the conversion rate above the existing **12%** rate.

QUESTION 5 — HOSPITAL AI-BASED SCHEDULING SYSTEM

Given

The hospital wants to determine whether the new AI system **reduces average waiting time**.

p-value:

0.08

Significance level:

5% = 0.05

Step 1: State the hypotheses

H0: The new system does not reduce average waiting time.

H1: The new system reduces average waiting time.

Step 2: Compare the p-value with the significance level

Given:

$$\mathbf{p\text{-}value = 0.08}$$

and:

$$\mathbf{\alpha = 0.05}$$

Compare:

$$\mathbf{0.08 > 0.05}$$

The p-value is greater than the significance level.

Step 3: Decision

Decision rule:

If $p\text{-}value > \alpha$, fail to reject H0.

Here:

$$\mathbf{0.08 > 0.05}$$

Therefore:

Decision: Fail to reject H0

Step 4: Conclusion

Since the p-value of **0.08** is greater than the significance level of **0.05**, there is not enough statistical evidence to conclude that the AI scheduling system reduces average waiting time.

This does not prove that the system has no effect. It means that the evidence is **not strong enough at the 5% significance level** to support the claim.

Final Answer

There is insufficient evidence to conclude that the AI system reduces waiting time.

Therefore, the hospital should not conclude that the new AI-based scheduling system reduces average waiting time at the 5% significance level.